REFERENCE DOCUMENT

Documentation

Detailed install, run, command, troubleshooting, and use-case guide

Team: TraversalAI

Members: Ahmad Alfaqeih, Mustafa Maisara

Coach: Ms. Amal Qutairi

CONTENTS

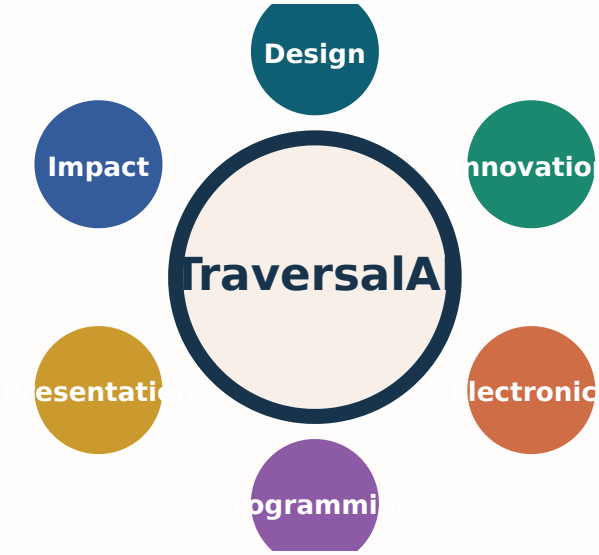
Table of Contents

Project Snapshot 3
Alignment With Judging Criteria 4
Install From Source 5
Install As a CLI Program 6
Starting the App 7
Software Platform 8
Hardware Platform 9
Program Command Catalog 10
Example Use Cases 11
Troubleshooting 12
Demo Readiness 13

SECTION

Project Snapshot

TraversalAI is a practical prototype that combines a chat interface, a central gateway, and specialized agents in one visible system. In the challenge context it can be directed at tourism and hospitality scenarios such as smart guidance, service coordination, and multi-channel assistance.



SECTION

Alignment With Judging Criteria

- Design: clarifies the problem, flow, and boundaries.
- Innovation: combines agents, tools, and skills in one coherent workflow.
- Electronics: uses Raspberry Pi 5 with microphone and speaker for a real AI + IoT setup.
- Programming: routes execution through the gateway, saved sessions, and model switching.
- Presentation: organized to support a short clear live explanation.
- Impact: ties the system to a real tourism or hospitality problem.

SECTION

Install From Source

```
git clone https://github.com/xplasma0/traversalai-v2.git
cd traversalai-v2
corepack pnpm install
corepack pnpm build
node traversalai.mjs --help
```

This is the right path for developers, judges reviewing the repository, or anyone validating the project locally before a global install.

SECTION

Install As a CLI Program

```
npm install -g .
traversalai --help
traversalai onboard
traversalai chat --no-open
```

After installation, the `traversalai` command becomes available system-wide. The normal first-run path is onboarding, then chat or gateway commands.

SECTION

Starting the App

```
traversalai onboard
traversalai gateway --force
traversalai chat --no-open
traversalai status
traversalai models --help
traversalai agents --help
```

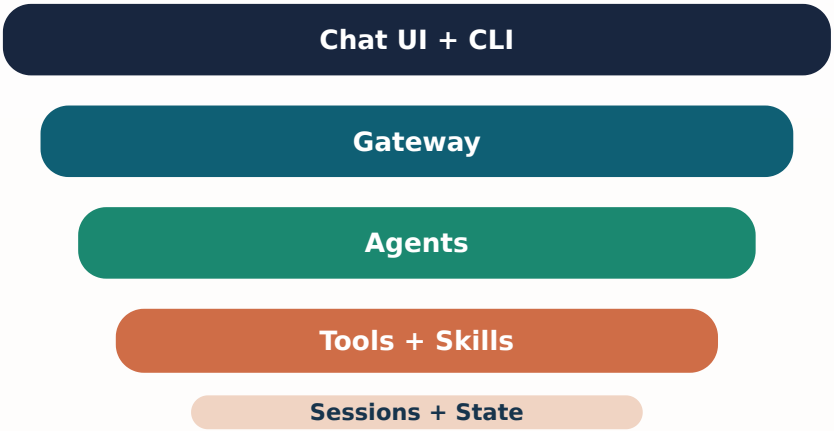
A typical startup sequence is: complete onboarding, start the gateway, open the chat interface, then inspect status, models, and agents.

Problem | Request | Decision | Execution | Outcome

SECTION

Software Platform

The gateway is the main integration boundary. Through it, the system switches agents and models, executes tools and skills, and preserves saved chats for later continuation.



SECTION

Hardware Platform

The prototype is designed to run locally on a Raspberry Pi 5 with microphone input and speaker output. That gives the project a concrete AI + IoT shape and makes it suitable for a live demonstration.

SECTION

Program Command Catalog

```
acp, agent, agents, approvals, browser, channels, chat, completion, config,
configure, cron, dashboard, devices, directory, dns, docs, doctor, gateway,
health,
hooks, logs, memory, message, models, node, nodes, onboard, pairing, plugins, qr,
reset, sandbox, secrets, security, sessions, setup, skills, status, system, tui,
uninstall, update, webhooks
```

These are the top-level commands exposed by the binary. Each one supports `--help` for more detail.

SECTION

Example Use Cases

- A tourism assistant that answers visitor questions before arrival, on site, and after the trip.
- An email agent that drafts, organizes, and tracks service responses.
- A social media agent that posts updates and monitors engagement channels.
- A Raspberry Pi 5 field kiosk with microphone and speaker for a live demonstration workflow.
- Saved sessions that let the operator resume the same task across multiple interactions.

SECTION

Troubleshooting

```
traversalai doctor
traversalai status
traversalai gateway --force
traversalai chat --no-open
traversalai config --help
```

- If the UI does not open correctly, confirm that the gateway is running and use the printed local URL.
- If models or agents are missing, inspect gateway setup and rerun `status` or `doctor`.
- If source installation fails, rerun `corepack pnpm install` followed by `corepack pnpm build`.
- If sessions behave unexpectedly, open a new chat and retest through the gateway-backed path.

SECTION

Demo Readiness

```
1. State the problem in under 30 seconds
2. Show the device or hosted runtime
3. Run one short scenario from request to outcome
4. Expose tool or skill activity during execution
5. Close with impact and scalability
```

This document uses guidebook_en_2025.pdf as a reference framework and does not replace it.